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30 May 1975
SUPERSEDING
MIL-W-21425A
23 April 1962

MILITARY SPECIFICATION

WIRE, STEEL, HIGH CARBON, SQUARE, UNCOATED, FOR MECHANICAL SPRINGS (GENERAL PURPOSE)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers three types of uncoated, square (0.050 to 0.500 inches in width), high-carbon steel wire (see 6.1).

1.2 **Classification.** Wire shall be of the following types, as specified (see 6.2):

Type I	Oil tempered spring wire for the manufacture of springs which require no subsequent hardening.
Type II	Hard drawn wire for the manufacture of springs which require no subsequent hardening.
Type III[1]	Soft (0.55 to 0.75 carbon) wire for the manufacture of springs which are to be hardened after cold forming, furnished in two conditions as follows (see 6.1.3): Condition A – Annealed at finished size Condition B – Annealed and lightly drawn.

[1] If a condition letter is not specified for type III wire, either condition A or condition B may be selected at the option of the spring manufacturer.

FSC 9505

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

STANDARDS

Federal

FED-STD-66	Steel: Chemical Composition and Hardenability
FED-STD-151	Metals; Test Methods

Military

*MIL-STD-129	Marking for Shipment and Storage
MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage

(Copies of specification, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

A 318	Tension Testing of Steel Spring Wire

(Application for copies should be addressed to the American Society for Testing and Materials, 1901 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 **Process.** The steel for the manufacture of the wire shall be made by the open-hearth, basic oxygen, or electric-furnace method.

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3.2 **Chemical composition.** The composition of the steel shall be as shown in table I, subject to the variations for check analysis given in table 4 of FED-STD-66.

Table I

CHEMICAL COMPOSITION (LADLE ANALYSIS)

Type	Carbon (%)	Manganese (%)	Silicon (%)	Phosphorus (max.) (%)	Sulfur (max.) (%)
I and III ...	0.55–0.75	0.60–1.20[1]	0.10–0.30	0.040	0.050
II	0.45–0.75[2]	0.60–1.20[3]	0.10–0.30	0.040	0.050

[1]For type III wire, manganese shall be between 0.80 and 1.20 percent for widths 3/16 inch and larger.
[2]Not varying more than 0.20 percent in any one lot.
[3]Not varying more than 0.30 percent in any one lot.

*3.3 **Defects.** The wire shall be free from cracks, pipes, and other detrimental imperfections (see 6.1).

*3.4 **Surface.** All wire shall be smooth and free from rust. Type I and type III, condition A, wire shall be free from excessive surface oxidation and shall be oiled or given an equivalent protective coating. Type II and type III, condition B, wire shall be free from scale or surface oxidation and shall be oiled.

3.5 **Uniformity of form.** All wire shall be sufficiently uniform in quality and temper to permit manufacture of springs. The wire shall not be wavy or kinked.

3.6 **Dimensions.** The wire shall be of the size specified, subject to the tolerances given in tables II and III.

3.7 **Wrapping properties.** The wire shall be free from cracks after wrapping around a mandrel of the size shown in table IV. Unless otherwise specified (see 6.4), the wrapping test shall not be required for wire of widths over 0.312 inch or for type III material.

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Table II

TOLERANCES FOR WIDTH AND OUT-OF-SQUARE
(All dimensions are in inches)

Width	Tolerance plus and minus	Out-of-square maximum
0.050 to 0.071, incl.	0.0015	0.0020
0.072 to 0.375, incl.	0.0030	0.0040
0.376 to 0.500, incl.	0.0045	0.0060

Table III

CORNER RADII
(All dimensions are in inches)

Width	Corner radius	
	Minimum	Maximum
0.050 to 0.125, incl.	0.012	0.017
0.126 to 0.188, incl.	0.020	0.025
0.189 to 0.250, incl.	0.025	0.035
0.251 to 0.312, incl.	0.030	0.040
0.313 to 0.500, incl.	0.035	0.045

Table IV

MANDREL SIZE FOR WRAPPING TEST
(All dimensions are in inches)

Wire width	Diameter of mandrel
0.050 to 0.162, incl.	Diagonal of wire
0.163 to 0.312, incl.	Twice the diagonal of wire

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3.8 **Mechanical properties.**

3.8.1 **Type I wire.** Type I wire shall be quenched and tempered to obtain the tensile strength specified in table V.

Table V

TENSILE STRENGTH, TYPE I WIRE

Width[1] (in.)	Tensile strength (psi)	
	Minimum	Maximum
0.050	253,000	283,000
0.054	247,000	277,000
0.062	241,000	271,000
0.072	235,000	265,000
0.080	230,000	260,000
0.091	225,000	255,000
0.105	220,000	250,000
0.120	215,000	240,000
0.135	210,000	235,000
0.148	205,000	230,000
0.162	200,000	225,000
0.177	195,000	220,000
0.192	190,000	215,000
0.207	188,000	213,000
0.225	185,000	210,000
0.250	183,000	208,000
0.312	180,000	205,000
0.375	175,000	200,000
0.437	170,000	195,000
0.500	165,000	190,000

[1] For widths other than those shown in table V, tensile strength shall be determined by interpolation.

3.8.2 **Type II wire.** Type II wire shall be cold drawn to obtain the tensile strength specified in table VI.

3.8.3 **Type III wire.**

3.8.3.1 **Condition A.** Type III, condition A, wire shall be annealed at finished size and shall have no cold work in the finished condition. The maximum tensile strength shall be 95,000 psi.

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Table VI

TENSILE STRENGTH, TYPE II WIRE

Width[1] (in.)	Tensile strength (psi)	
	Minimum	Maximum
0.050	243,000	279,000
0.054	237,000	272,000
0.062	232,000	266,000
0.072	227,000	261,000
0.080	221,000	254,000
0.091	216,000	248,000
0.105	210,000	241,000
0.120	206,000	237,000
0.135	203,000	234,000
0.148	200,000	230,000
0.162	195,000	225,000
0.177	192,000	221,000
0.192	190,000	218,000
0.207	186,000	214,000
0.225	182,000	210,000
0.250	174,000	200,000
0.312	167,000	193,000
0.375	162,000	187,000
0.437	156,000	180,000
0.500	152,000	176,000

[1] For widths other than those shown in table VI, tensile strength shall be determined by interpolation.

*3.8.3.2 **Condition B.** Type III, condition B, wire shall be annealed and shall be given at least one pass through dies or rolls after the last annealing treatment. The minimum tensile strength shall be 100,000 psi and the maximum, 130,000 psi.

3.8.3.3 **When heattreated.** Type III wire, when heattreated, shall develop the properties of table V.

*3.9 **Cast.** Type II wire shall be properly cast. When tested as specified in 4.5.4, wire shall lie substantially flat on itself and shall not spring up and show a wavy condition.

3.10 **Coil sizes and weights.** Unless otherwise specified, wire shall be furnished in the coil diameters and weights specified in table VII for the appropriate wire size.

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Table VII

APPROXIMATE COIL DIAMETERS AND WEIGHTS

Wire type	Wire size (in.)	Approx. mean coil diameter (in.)	Approx. coil weights (lb)
I	0.050-0.080	24, 22, 16	75-300
	0.081-0.160	54, 48, 42, 36, 30	125-600
	0.161-0.375	60, 54, 48, 42, 36	150-600
	0.376-0.500	72, 60, 54	150-600
II and III	0.050-0.0625	22, 16	75-300
	0.063-0.1875	30, 28, 26, 24, 22	125-600
	0.188-0.375	36, 30, 28, 26, 24, 22	150-600
	0.376-0.500	36, 30 28, 26	150-600

3.11 **Identification marking.** Each coil shall have two substantial tags, printed with the name or trademark of the manufacturer, the specification number, the size, the type, and the condition.

*3.12 **Workmanship.** The wire shall be substantially uniform in quality and temper and shall not have imperfections of a nature or degree, for the grade and quality ordered, that will be detrimental to the forming or fabrication of finished parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Lot.** A lot shall consist of wire of the same size, type, condition, and composition, manufactured from the same heat of steel.

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4.3 **Sampling.**

*4.3.1 **Sampling for chemical tests.** When specified, three samples shall be taken from each heat for chemical tests. Specimens for chemical analysis shall be in the form of wire clippings, free from oil, dirt, or other foreign material and shall be representative of the full cross section of the wire. (It is preferable that the product check analysis be performed earlier, such as in the rod or billet stage.)

*4.3.2 **Sampling for other tests.** Five sample coils shall be selected at random from each lot. Specimens shall be taken from one end of each sample coil for tensile, cast (type II and type III, condition B), and wrapping tests.

4.4 **Inspection.**

4.4.1 **Contractor's inspection.** The contractor shall determine that all wire offered for Government acceptance or delivery conforms to the requirements of the contract and to all applicable documents.

4.5 **Tests.**

4.5.1 **Chemical tests.** Chemical tests shall be made in accordance with FED-STD-151, method 111.2. The samples shall be analyzed for conformance to 3.2.

4.5.2 **Tensile test.** Tensile strength tests shall be made in accordance with ASTM A 318. The test specimen shall be straight, and the length of the specimen between the jaws of the testing machine shall be not less than 10 inches. Unless otherwise specified, the crosshead speed of the testing machine shall be not less than 1/4 inch per minute and not more than 5 inches per minute.

4.5.3 **Wrapping test.** Wire small enough to be wrapped by hand may be tested by bending one end of the wire over itself to form a loop with ends crossing at right angles, the shorter end being several inches long, and then tightly winding the longer end five complete turns about the other with turns close together. For larger wire or optionally for wire of any size, the wrapping may be done by any hand or power device which will wrap the wire closely about a mandrel of the required diameter (see 3.7) for five complete turns. The sample shall be considered to have failed if any cracks occur in the wire after the first complete turn. A test shall be repeated if a crack occurs in the first turn (since the wire may be bent locally to a radius less than that specified).

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*4.5.4 **Cast test (type II and type III, condition B).** A sample consisting of approximately two full turns of wire carefully cut and removed from a coil shall be placed on a flat surface and examined for conformance to 3.9.

5. PREPARATION FOR DELIVERY

5.1 **Levels of packaging and packing.** Material shall be prepared for shipment at the levels specified in procurement documents (see 6.2).

5.1.1 **Level A.** Level A packaging and packing shall be in accordance with MIL-STD-163.

5.1.2 **Level C.** Level C packaging and packing shall be in accordance with commercial practice to insure carrier acceptance for safe delivery at the destination at the lowest rate. As a minimum, the shipment shall meet the requirements of carrier rules and regulations applicable to the mode of transportation.

*5.2 **Marking.** In addition to any marking required by the contract or order, marking shall be in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 **Intended use.** This specification covers general purpose carbon steel spring wire intended for springs not subjected to severe repeated stress. Other quality wire, such as music spring wire or valve spring quality wire, should be used where freedom from surface defects and decarburization are required to insure maximum fatigue life for springs operating under repetitive stress.

6.1.1 **Type I.** The mechanical properties of type I wire are obtained by heat treatment. Springs formed from type I wire are used without further heat treatment, except for suitable low-temperature treatment to relieve residual stresses caused by coiling. Springs formed from type I wire can generally be held to closer dimensional tolerances than springs formed from type II wire.

6.1.2 **Type II.** The mechanical properties of type II wire are obtained by cold working. Springs formed from type II wire are used without further heat treatment, except for suitable low-temperature treatment to relieve residual stresses caused by coiling.

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6.1.3 **Type III.** Type III wire is a soft, spheroidized annealed wire used where springs must be formed of soft wire. Type III, condition A, wire is generally suitable for more severe deformation than type III, condition B, wire. Springs should be heattreated to specified mechanical properties after forming. The composition of type III is the same as type I.

6.2 **Ordering data.** Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Type of wire
c. For type III wire, the condition (see 1.2).
d. Width of wire.
e. Quantity of wire.
f. Coil size and weight of wire where desired (see 3.10)
g. Level of packaging (see section 5).

6.3 **Transportation information.** Transportation information applicable to this item is as follows:

Wire, steel, plain
Carload minimum weight 40,000 pounds
Motor volume minimum weight 36,000 pounds.

6.4 **Wrapping test.** Wire larger than 0.312 inches in width should withstand wrapping around a mandrel equivalent to three times the nominal diagonal. This test is generally not specified, however, because of the difficulty in conducting the test with wrapping equipment normally available.

*6.5 **Changes from previous issue.** The margins of this specification are marked with asterisks to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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